

Walking in Space: Immersive Storytelling in Space Exploration VR Museum

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Figure 1: A VR storytelling experience focuses on humanity's history of space exploration within a museum setting. Impossible spaces were utilized to enable seamless walking navigation throughout the virtual museum.

ABSTRACT

Virtual Reality (VR) is widely used in museums to offer unique exhibition experiences. Navigation in virtual museums is typically achieved through teleportation. While teleportation bypasses physical space limitations, this technique differs from natural walking in the real world. Walking provides more immersive experiences but usually requires a large physical area. To address this, we present a space exploration VR museum where users can walk through the history of humanity's space exploration within just a 2.5 x 2.5 m physical space by employing overlapping spaces. This approach enables immersive storytelling and accessible virtual museum experiences, with preliminary results supporting its effectiveness.

Index Terms: Virtual Reality, Walking, Impossible Spaces, Virtual Museum.

1 INTRODUCTION

Virtual Reality (VR) has become a powerful tool for immersive and interactive museum experiences, allowing users to engage with exhibits and explore content in novel ways. However, navigation in VR can significantly impact the user experience. While teleportation supports exploration in small physical spaces, it can limit the sense of immersion and presence [2]. Conversely, natural walking enhances immersion but requires significant physical space.

Techniques such as redirected walking [4] and impossible spaces [8] address these challenges by enabling walking in VR within smaller physical areas. Some virtual museums have adopted these methods, combining the immersive benefits of walking with the practicality of limited space [10].

This paper presents a virtual museum experience on space exploration (Figure 1), utilizing impossible space techniques to allow

walking within a 2.5 x 2.5 m area. Preliminary user feedback highlights the potential of this approach to deliver an immersive, engaging museum experience accessible from home.

2 RELATED WORK

In museums, VR offers opportunities to reconstruct historical environments and create engaging, immersive experiences [7].

Locomotion in Virtual Reality: VR offers various navigation methods, with teleportation being common. *Teleportation* involves pointing to a virtual location and instantly moving there, making it ideal for exploring large spaces in small physical areas [5]. While efficient, teleportation is less natural than walking, which feels more natural and engaging [2]. *Natural Walking*, mirroring the user's movements, has been pointed as a more immersive technique [2] but requires significant space for the user to move. Techniques such as redirected walking and impossible spaces address this limitation by manipulating users' perceptions to maximize the physical space. *Redirected walking* [4] subtly steers users away from boundaries through unnoticed perceptual gains. *Impossible spaces* [8] use non-Euclidean spaces, maximizing space with self-overlapping layouts. Portals and corridors [6] leverage this overlap to connect and merge virtual rooms or create seamless paths between them.

Virtual Reality in Museums: VR has a wide range of applications in museums, offering diverse use cases and experiences. Wang et al. [9] used teleportation and developed tools such as a 2D mini-map and the World-in-Miniature metaphor to enhance navigation in VR museums and improve user experience. Lugin et al. [3] examined a location-based VR museum in a large shared physical space (600m²), enabling multiple users to walk and explore together. Yu et al. [10] applied redirected walking to create a World War I (WWI) narrative in VR, where users walked in a 4.5 x 1.5 m space simulating WWI tunnels, immersing themselves in the soldier experience. While these works incorporate walking, they require much larger areas than the space VR users typically have available at home, which average around 2.4 x 2.2 m [1]. Our use case presents a virtual museum storytelling experience, enabling users to walk within a small 2.5 x 2.5 m space.

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3 WALKING IN SPACE

This work presents a VR storytelling experience designed to enable walking in small spaces (2.5 x 2.5 m) by employing overlapping spaces, allowing users to enjoy the experience in their homes or offices. The virtual museum focuses on delivering an immersive narrative about humanity's history of space exploration.

Three phases shaped the museum experience, as shown in Figure 1. The first five rooms, representing the *Medieval Era*, depicted astrology's central role in society and showcased ideas such as the Ptolemaic model of the solar system. The next five rooms, from the *Modern Era*, chronicled the shift to a scientific worldview, starting with Copernicus's heliocentric model and highlighting key discoveries. This phase guided users through humanity's journey to the moon, including an interactive moon base experience. The final five rooms represented the *Future Era* and explored the solar system's planets, moons, and unique features, concluding with a focus on black holes and related discoveries. The progression gradually immersed users in space exploration history, starting with traditional museum rooms, moving to open-roof rooms for stargazing, and culminating in a spaceship visit with close-up views of planets.

Throughout the museum, every room contained artifacts with descriptive information. Some artifacts were interactive, and users could also collect and play music discs to enjoy during their visit. To facilitate navigation between rooms and ensure users could walk freely without crossing physical boundaries, impossible space techniques, such as portals and corridors, were implemented. *Portals* previews the next room and overlaps with the current room, allowing users to quickly transition between spaces. The portals, adapted from *PortalsVR* by Lochner et al. [2], were carefully positioned to connect two rooms, ensuring alignment to avoid introducing offsets in the VE. *Corridor* entrances are embedded in the virtual walls and lead users to a transition area, enabling them to walk comfortably between rooms without noticing the overlap between the rooms and the corridor. These corridors were designed by generating a path connecting the entry doors of two rooms while ensuring the user remains within the physical boundaries. The corridor was then constructed along this path to seamlessly link the rooms. These techniques seamlessly connected and overlapped the museum rooms, allowing users to walk between them within the constraints of the physical area. The entire experience lasted around 30 minutes.

3.1 Preliminary Results

A user study with 24 participants from diverse backgrounds was conducted in a laboratory with a 2.5 x 2.5 m tracking area to evaluate the virtual museum experience. Participants used an HTC VIVE Cosmos headset and one controller to interact with the artifacts. The user study assessed presence, cybersickness, spatial awareness, and overall user experience.

Preliminary results indicated that users enjoyed the experience and felt a strong sense of immersion and presence in the virtual museum. One participant noted, "Even though I know I will not fall off, it still gives me that feeling," when describing their experience in the black hole-themed rooms. Participants also found the navigation comfortable within the virtual museum, despite being walking in a small physical space. As one participant noted, "Walking between rooms through the corridor is natural, comfortable, and fun." The techniques used to connect rooms — portals and corridors — showed a positive impact on navigation. Most participants preferred using corridors for their intuitiveness and ease, as one remarked, "Corridors were intuitive and easy to not get lost." Others favored portals for their practicality in quickly transitioning between rooms and previewing the next room, with one participant commenting, "I liked portals more because of their practicality."

Participants were asked to rank the order of which rooms they visited chronologically to gauge if they successfully followed the museum's narrative and understood the chronological order of

events. Around 60% of the participants could follow along with the narrative even with the virtual museum's impossible spatial design.

Most participants felt they learned something new and found space exploration captivating. Several expressed interest in re-researching the topic further, with one participant stating, "It would be more interesting to learn about history in school this way in VR."

4 CONCLUSION

This paper presents a VR museum storytelling experience where users can explore a narrative about humankind's space exploration while walking in a small physical space. Our approach employs impossible space techniques, such as portals and corridors, to enable walking within a small physical area of just 2.5 x 2.5 m. This method allows users to navigate seamlessly through virtual museum rooms without crossing physical boundaries, enabling them to walk at their own pace and making the experience accessible in smaller spaces. Preliminary user feedback revealed positive levels of immersion, natural movement, and engagement with the virtual museum's content. Notably, the impossible spaces used in the virtual museum did not show evidence of diminishing the users' understanding of the narrative or the chronological order of historical events. These results underscore the potential of impossible space techniques to enhance navigation in storytelling experiences, improve user experience, and make VR museum applications more accessible for small physical spaces.

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